

Transmission — Four Wheel Drive (4WD)

 [Ford Ecat Electronic Parts Catalog](#)

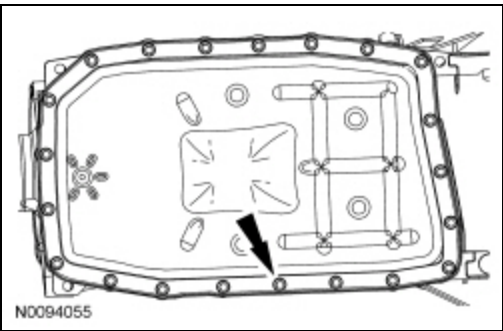
Special Tool(s)	
 ST1636-A	Retainer, Torque Converter 307-346 (T97T-7902-A)

Removal

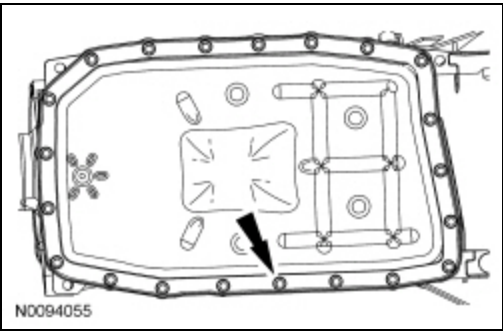
NOTE: [Refer to the Transmission Cooler Flushing Job Aid on the FMCDealer website.](#)

All vehicles

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to Section 100-02.
2. Disconnect the battery ground cable. For additional information, refer to Section 414-01.
3. If transmission disassembly or installation of a new transmission is necessary, remove the 21 transmission fluid pan bolts and remove the transmission fluid pan. Drain the transmission fluid into a container.



4. Install the transmission fluid pan and the 21 transmission fluid pan bolts. Tighten transmission fluid pan bolts in a crisscross pattern.
 - Tighten to 12 Nm (106 lb-in).



5. Remove the exhaust Y-pipe. For additional information, refer to Section 309-00.

6. **NOTE:** *It is only necessary to remove the transfer case from the transmission if a new transmission is being installed or if the transmission is going to be disassembled for a transmission related repair.*

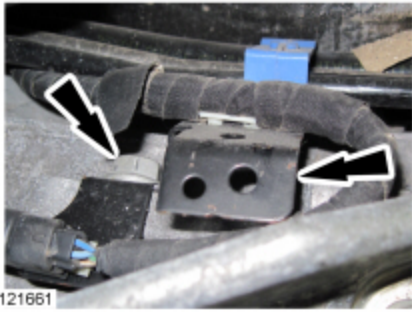
Remove the transfer case. For additional information, refer to Section 308-07B.

7. **NOTE:** *Make sure the transmission jack contacts the outer ribs of the transmission fluid pan.*

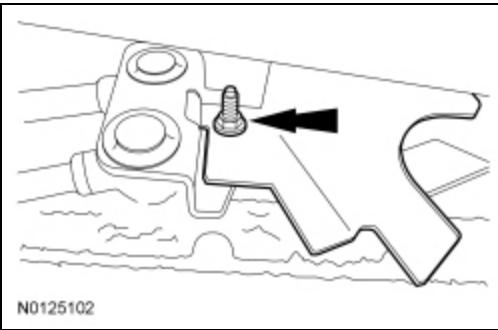
NOTE: *Make sure the transmission is securely fastened to the transmission jack.*

Using a high-lift transmission jack, support the transmission and remove the jack stand.

8. Remove the fuel line bracket bolt from the left side of the transmission and position the fuel line bracket aside.



9. If equipped, remove the nut and the selector lever cable splash shield.

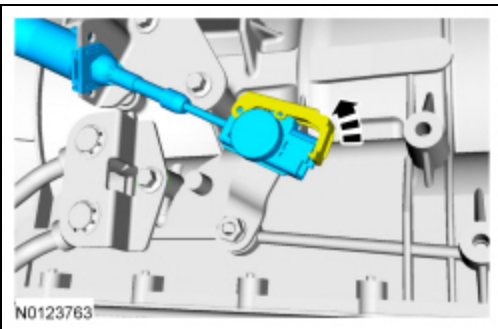


Vehicles equipped with a floor shift

10. **NOTICE:** **To prevent selector lever cable damage, do not apply force to the selector lever cable between the manual control lever and the selector lever cable bracket.**

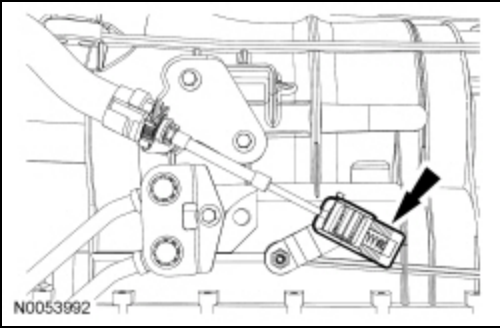
NOTE: *The 3.5L engine is shown, the 3.7L, 5.0L and 6.2L engines are similar.*

Rotate the cable adjustment lock upward and remove the selector lever cable from the manual control lever.



Vehicles equipped with a column shift

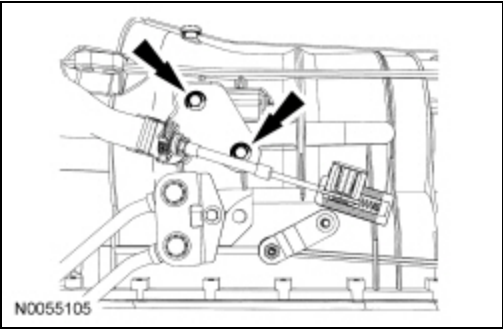
11. Move the locking tab up and disconnect the selector lever cable from the manual control lever.



All vehicles

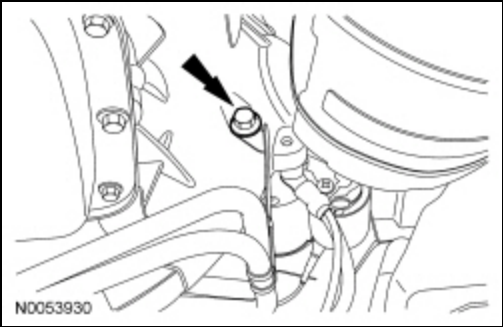
12. **NOTE:** *The column shift is shown, the floor shift is similar.*

Remove the selector lever cable bracket bolts and remove the selector lever cable bracket.



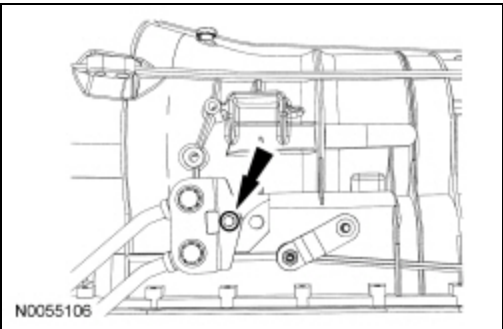
Vehicles equipped with a 5.0L or 6.2L engine

13. Remove the transmission-to-engine bolt.



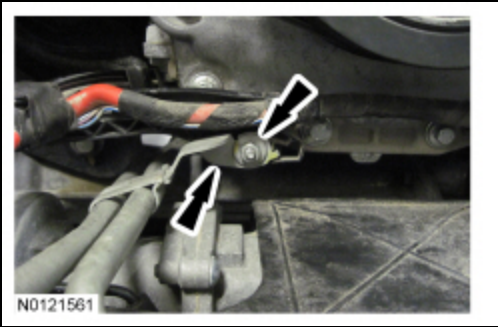
All vehicles

14. Remove and discard the transmission fluid cooler tube bracket bolt.



15. **NOTE:** *The 3.5L and 3.7L engine is shown, the 5.0L and the 6.2L engines are similar.*

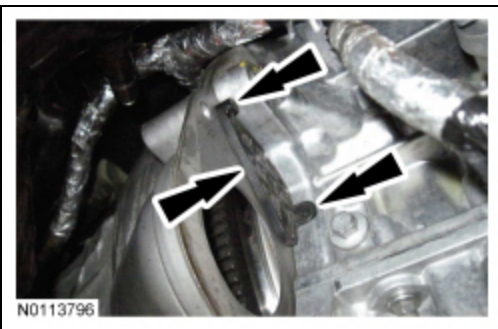
Remove the transmission fluid cooler tube bracket nut and position the transmission fluid cooler tube bracket and tubes aside.



16. Remove the starter motor. For additional information, refer to Section 303-06.

Vehicles equipped with a 3.5L or 3.7L engine

17. Remove the 2 pin-type retainers and the access cover.

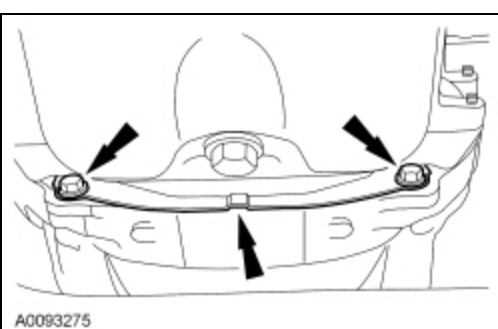


18. Remove and discard the 4 flexplate-to-torque converter nuts.

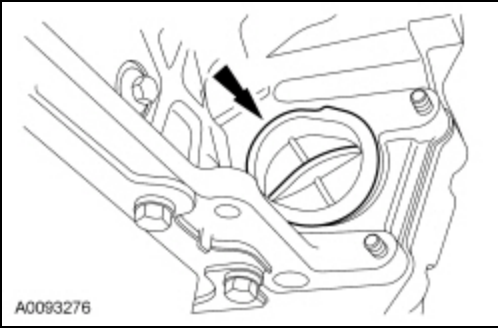


Vehicles equipped with a 5.0L or 6.2L engine

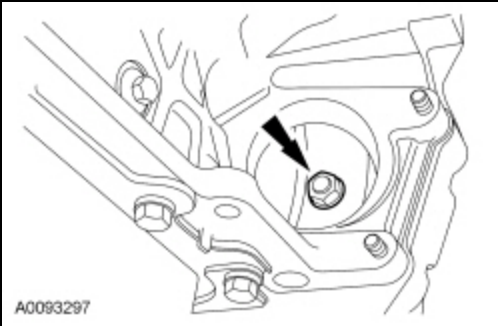
19. Remove the flexplate inspection cover bolts and the flexplate inspection cover.



20. Remove the rubber torque converter nut access cover.



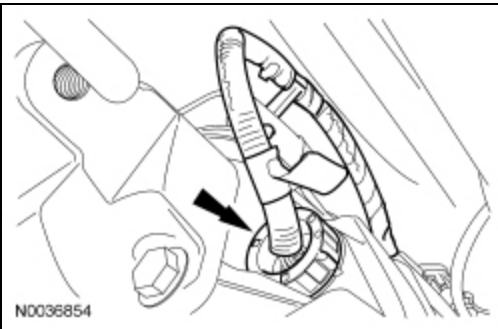
21. Remove and discard the 4 flexplate-to-torque converter nuts.



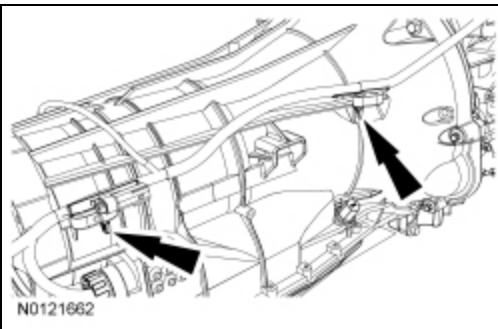
All vehicles

22. **NOTICE:** Do not pull on the wiring harness to disconnect the electrical connector or damage to the electrical connector will occur.

Disconnect the electrical connector by twisting the outer shell and pulling back on the electrical connector.



23. Disconnect the wiring harness retainers from the top of the transmission.



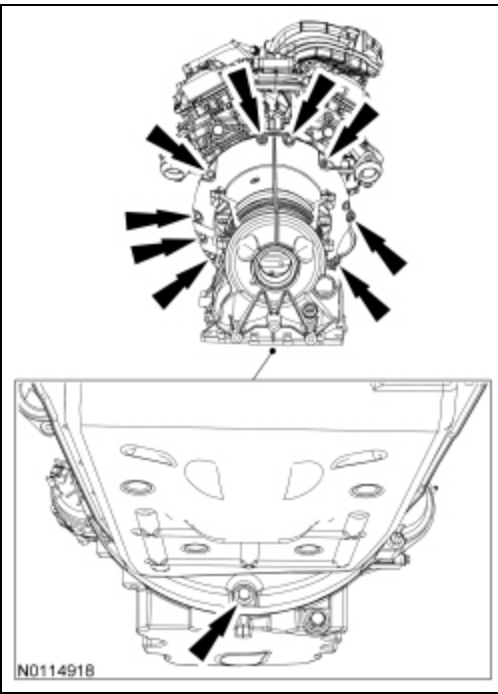
Vehicles equipped with a 3.5L or 3.7L engine

24. **WARNING:** Always secure transmission, transfer case, and axle assemblies to their service jack. Avoid obstructions while lowering and raising the jack. Improperly secured assemblies or contact with obstructions may cause the assembly to fall off the jack, which could result in serious personal injury.

NOTICE: Make sure to not pinch the transmission wiring harness between the transmission and the body or damage can occur.

NOTE: The top 2 transmission-to-engine bolts secure the fuel line bracket to the transmission case.

Remove the 10 transmission-to-engine bolts.



Vehicles equipped with a 5.0L or 6.2L engine

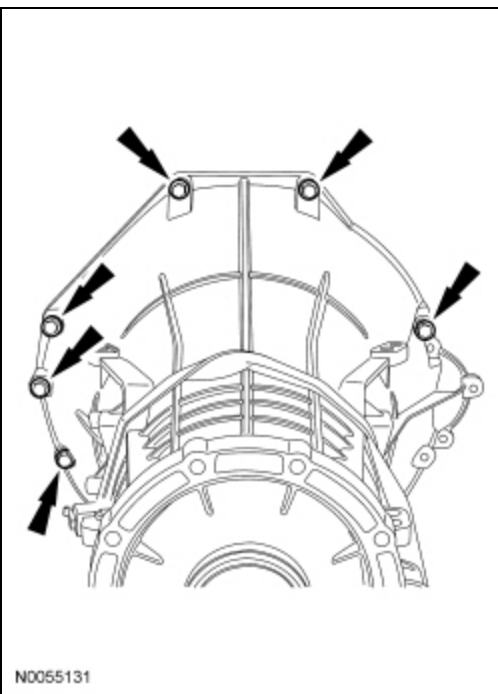
25. WARNING: Always secure transmission, transfer case, and axle assemblies to their service jack. Avoid obstructions while lowering and raising the jack. Improperly secured assemblies or contact with obstructions may cause the assembly to fall off the jack, which could result in serious personal injury.

NOTICE: Make sure to not pinch the transmission wiring harness between the transmission and the body or damage can occur.

NOTE: Remove the top 2 transmission-to-engine bolts before removing the rest of the bolts.

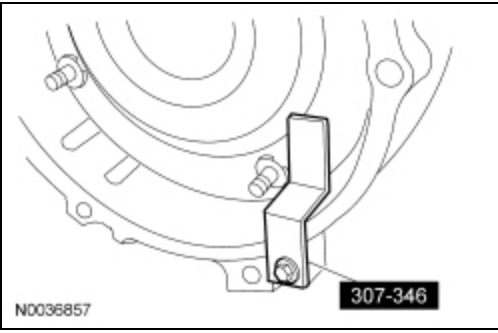
NOTE: The top left transmission-to-engine bolt secures the fuel line bracket to the transmission case.

Remove the 6 remaining transmission-to-engine bolts.



All vehicles

26. Slide the transmission back enough to install the Torque Converter Retainer.



27. Remove the transmission from the vehicle.

28. If the transmission is being disassembled to install new parts, or if a new or remanufactured transmission is being installed, clean and backflush the transmission fluid cooler, the auxiliary transmission fluid cooler (if equipped) and the transmission fluid cooler tubes to prevent contaminants from entering the transmission. For additional information, refer to Transmission Fluid Cooler Backflushing and Cleaning in this section.